

ASSIGNMENT-1

OPERATORS AND CONTROL STATEMENTS & LOOPING

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CSA0901

1.ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

```
1 import java.util.Scanner;
2 public class ar{
3     public static void main(String args[]){
4         Scanner arith = new Scanner (System.in);
5         int a= arith.nextInt();
6         int b= arith.nextInt();
7         System.out.println(a+b);
8         System.out.println(a-b);
9         System.out.println(a*b);
10        System.out.println(a/b);
11    }
12 }
13
```

INPUT

20

10

OUTPUT

30

10

200

2

2.MAX

Write a Java program to find the maximum of three numbers using conditional (if-else) statements.

```
1 public class maxx{
2     public static void main (String args []){
3         int a=10;
4         int b=20;
5         int c=30;
6         int maxi=0;
7         if (a>=b && a>=c){
8             maxi= a;
9
10        } else if (b>=a && b>=c){
11            maxi= b;
12        }else if(c>=a && c>=b){
13            maxi=c;
14        }
15        System.out.println("max number is: "+maxi);
16    }
17 }
```

INPUT

OUTPUT

30

3.LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

```
import java.util.Scanner;
public class leap{
    public static void main(String args[]){
        Scanner le = new Scanner(System.in);
        int yr= le.nextInt();
        if(yr % 4==0 && yr % 10!=0){
            System.out.println("the year is leap year");
        }else{
            System.out.println("the year is not leap");
        }
    }
}
```

INPUT

2024

OUTPUT

The year
is leap
year

4.FIBONACCI

Write a Java program to print the Fibonacci series up to a given number using loops.

```
1 import java.util.Scanner;
2 public class fibo{
3     public static void main(String args[]){
4         Scanner scan = new Scanner(System.in) ;
5         int n = scan.nextInt() ;
6         int a = 0;
7         int b = 1;
8         int x = 0;
9         while (x<n){
10             System.out.println(a);
11             int c = a+b;
12             a=b;
13             b=c;
14             x++;
15         }
16
17     }
18 }
```

INPUT

6

OUTPUT

0
1
1
2
3
5

5.FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

```
1 import java.util.Scanner;
2 public class fac{
3     public static void main(String args[]){
4         Scanner fact = new Scanner(System.in);
5         int n = fact.nextInt();
6         int x =1;
7         if(n<0){
8             System.out.println("the num is invalid!!");
9         }else
10        {
11            while (n>0){
12                x=x*n;
13                n-=1;
14            }
15        }
16        System.out.println(x);
17    }
18 }
```

INPUT

3

OUTPUT

6

6.RT-PATTERN

Write a Java program to print the following pattern using loops:

*
**


```
1 public class maxx{
2     public static void main (String args []){
3         int a=10;
4         int b=20;
5         int c=30;
6         int maxi=0;
7         if (a>=b && a>=c){
8             maxi= a;
9         }
10        } else if (b>=a && b>=c){
11            maxi= b;
12        }else if(c>=a && c>=b){
13            maxi=c;
14        }
15        System.out.println("max number is: "+maxi);
16    }
17 }
```

INPUT

5

OUTPUT

*

**

7.SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int a[] = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        int x = sc.nextInt();
14
15        int l = 0, h = n - 1;
16
17        while (l <= h) {
18            int m = (l + h) / 2;
19
20            if (a[m] == x) {
21                System.out.println("Found");
22                return;
23            } else if (a[m] < x)
24                l = m + 1;
25            else
26                h = m - 1;
27        }
28
29        System.out.println("Not Found");
30    }
31 }
```

INPUT

3

10 30 20

10

OUTPUT

FOUND

8.GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class gcd{
3     public static void main (String args[]){
4         Scanner scan = new Scanner(System.in);
5         int a = scan.nextInt();
6         int b= scan.nextInt();
7         int min = Math.min(a,b);
8         while(true){
9             if (a%min==0 && b%min==0){
10                 System.out .println(min);
11                 break;
12             }
13             min--;
14         }
15     }
16 }
```

INPUT

48

18

OUTPUT

6

9.PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class prime{
3     public static void main(String args[]){
4         Scanner pr = new Scanner(System.in);
5         int a= pr.nextInt();
6         int b= 0;
7         for(int i=2; i<a; i++){
8             if (i%a==0){
9                 b= 1;
10            }
11        }
12        if (b==1){
13            System.out.println("the num is not prime");
14        }
15        else{
16            System.out.println("the num is prime");
17        }
18
19
20    }
21 }
```

INPUT

11

OUTPUT

The num
is prime

10.SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class sort{
3     public static void main(String args[]){
4         Scanner scan = new Scanner(System.in);
5         int n = scan.nextInt();
6         int[] a= new int[n];
7         for (int i=0; i<n;i++){
8             a[i]= scan.nextInt();
9         }
10        for (int i=0;i<n-1; i++){
11            int min =i;
12            for (int j=i+1;j<n;j++){
13                if(a[j]<a[min]) {
14                    min=j;
15                }
16            }
17            int temp=a[i];
18            a[i]=a[min];
19            a[min]=temp;
20        }
21        for (int i=0;i<n;i++){
22            System.out.println(a[i]+" ");
23        }
24    }
25 }
```

INPUT

3

12 10 11

OUTPUT

10

11

12

11. TABLE

Write a Java program to print the multiplication table of a given number using loops.

```
1 import java.util.Scanner;
2 public class table{
3     public static void main(String args[]) {
4         Scanner tb = new Scanner (System.in);
5         int a = tb.nextInt();
6         for (int i = 1; i<=10; i++ ){
7             System.out.println("2*" + i + "=" +a*i);
8         }
9     }
10 }
```

INPUT

2

OUTPUT

2*1=2
2*2=4
2*3=6
2*4=8
2*5=10
2*6=12
2*7=14
2*8=16
2*9=18
2*10=20

12. FACTORIAL

Write a Java program to find the factorial of a given number using recursion.

```
1 import java.util.Scanner;
2 public class fac{
3     static int fact(int n){
4         if (n==0 || n==1)
5             return 1;
6         else
7             return n*=fact(n-1);
8     }
9     public static void main (String args []){
10         Scanner scan = new Scanner(System.in);
11         int n = scan.nextInt();
12         System.out.println(fact(n));
13     }
14 }
```

INPUT

5

OUTPUT

120

13.LT-PATTERN

Write a Java program to print the following pattern using loops:

**

*

```
1 import java.util.Scanner;  
2 public class lt{  
3     public static void main(String args []){  
4         Scanner scan = new Scanner(System.in) ;  
5         int n= scan.nextInt();  
6         for (int i = n; i>=1; i--){  
7             for (int j = 1; j<=i; j++){  
8                 System.out.print("*");  
9             }  
10            System.out.println();  
11        }  
12  
13    }  
14 }
```

INPUT

5

OUTPUT

**

*

14.LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

```
1 import java.util.Scanner;  
2 public class lcm{  
3     public static void main(String args []){  
4         Scanner scan = new Scanner(System.in);  
5         int a = scan.nextInt();  
6         int b = scan.nextInt();  
7         int max = Math.max(a,b);  
8         while (true){  
9             if (max%a==0 && max%b==0){  
10                 System.out.println(max);  
11                 break;  
12             }  
13             max++;  
14         }  
15  
16     }  
17 }
```

INPUT

4
6

OUTPUT

12

15. D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class b_d{
3     public static void main(String args[]){
4         Scanner scan = new Scanner(System.in);
5         int n = scan.nextInt();
6         int b = 0,p=1;
7         while(n>0){
8             int r = n%2;
9             b=b+r*p;
10            p*=10;
11            n=n/2;
12        }
13        System.out.println(b);
14    }
15 }
```

INPUT	OUTPUT
10	1010

16. B TO D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class bd{
3     public static void main(String args[]){
4         Scanner scan = new Scanner(System.in);
5         int bi= scan.nextInt() ;
6         int d=0,b=1;
7         while(b>0){
8             int r= bi%10;
9             d=d+r*b;
10            b*=2;
11            bi/=10;
12        }
13        System.out.println(d);
14    }
15 }
```

INPUT	OUTPUT
1010	10

17.SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

```
1 public class sum{
2     public static void main(String args []){
3         int a= 4355;
4         int b=0;
5         while (a>0){
6             int x= a%10;
7             b=b+x;
8             a=a/10;
9         }
10
11     System.out.println(b) ;
12 }
13 }
```

INPUT

OUTPUT

17

18.PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

```
public class pro{
    public static void main(String args []){
        int a = 1234;
        int b=1;
        while (a>0){
            int x= a%10;
            b=b*x;
            a=a/10;
        }
        System.out.println(b);
    }
}
```

INPUT

OUTPUT

24

19.POWER

Write a Java program to find the power of a number using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class pow{
3     public static void main(String args[]){
4         Scanner scan = new Scanner(System.in);
5         int b = scan.nextInt();
6         int e = scan.nextInt();
7         int x = 1;
8         for(int i= 1; i<=e; i++){
9             x*=b;
10        }
11        System.out.println(x);
12    }
13 }
```

```
INPUT
2
2
```

```
OUTPUT
4
```

20.AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

```
1 import java.util.Scanner;
2 public class area{
3     public static void main(String args []){
4         Scanner scan = new Scanner (System.in);
5         double a= scan.nextInt();
6         double pi= 3.14;
7         double ar= pi*a*a;
8         System.out.println(ar);
9     }
10 }
```

```
INPUT
2
```

```
OUTPUT
12.56
```


21.SPHERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

```
1 import java.util.Scanner;
2 public class vol{
3     public static void main (String args []){
4         Scanner scan = new Scanner (System.in);
5         double a= scan.nextInt();
6         double b=1.333;
7         double c= 3.14;
8         double d = b*c*a*a*a;
9         System.out.println(d);
10    }
11 }
```

INPUT

4

OUTPUT

267.87968

22. PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class pali{
3     public static void main(String args[]){
4         Scanner pal = new Scanner(System.in);
5         String a= pal.nextLine();
6         String b= "";
7         for(int i=a.length() - 1 ; i>=0; i--){
8             b=b+a.charAt(i);
9         }if (b.equals(a)){
10             System.out.println("its a palindrome!!!");
11         }else{
12             System.out.println("its not a palindrome :(");
13         }
14     }
15 }
16 }
```

INPUT

noon

OUTPUT

Its a
palindrome!!!

23. AMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class ams{
3     public static void main(String args []) {
4         Scanner scan = new Scanner(System.in);
5         int a = scan.nextInt();
6         String snum= Integer.toString(a);
7         int l = snum.length();
8         int x=0;
9         for(int i= 0 ; i<snum.length(); i++){
10             int digit= Character.getNumericValue(snum.charAt(i));
11             int pow=(int) Math.pow(digit,l);
12             x=x+pow;
13         }
14         System.out.println(x);
15     }
16 }
17 }
```

INPUT

213

OUTPUT

36

24.RT-NUM.PATTERN

Write a Java program to print the following pattern using loops:

```
1
22
333
4444
55555
```

```
1 import java.util.Scanner;
2 public class rtnum{
3     public static void main(String args []){
4         Scanner scan = new Scanner(System.in);
5         int n = scan.nextInt();
6         for (int i = 1; i<=n; i++){
7             for (int j = 1; j<=i; j++){
8                 System.out.print(i);
9             }
10            System.out.println();
11        }
12
13    }
14 }
```

INPUT

5

OUTPUT

1

22

333

4444

55555

25.PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class per{
3     public static void main(String args[]){
4         Scanner scan = new Scanner(System.in);
5         int a= scan.nextInt();
6         int sum = 0;
7         for(int i = 1; i<a; i++){
8             if (a%i==0)
9                 sum+=i;
10
11         }
12         if (a==sum)
13             System.out.print("its a perfect num");
14         else
15             System.out.println("its not a perfect num") ;
16     }
17 }
```

INPUT

6

OUTPUT

Its a perfect num

26.MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class max{
3     public static void main(String args[]){
4         Scanner sc= new Scanner(System.in);
5         int n = sc.nextInt();
6         int [] a= new int[n];
7
8         for (int i =0;i<n;i++)
9             a[i]=sc.nextInt();
10        int max=a[0];
11
12        for (int i=1; i<n;i++){
13            if(a[i]>max)
14                max=a[i];
15
16        }
17
18        System.out.println(max);
19    }
20 }
```

INPUT

5

10 50 20 40 30

OUTPUT

50

27. 2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class secmax{
3     public static void main(String args[]){
4         Scanner sc= new Scanner(System.in);
5         int n= sc.nextInt();
6         int [] a= new int[n];
7         for (int i=0; i<n;i++){
8             a[i]=sc.nextInt();
9         }
10        int large = a[0];
11        int sec = a[0];
12        for (int i=1; i<n;i++){
13            if(a[i]>large){
14                sec=large;
15                large=a[i];
16            }
17            else if (a[i]>sec && a[i]!=large){
18                sec=a[i];
19            }
20        }
21        System.out.println(sec);
22    }
23 }
```

INPUT

5

10 20 30 40

50

OUTPUT

40

28.MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class min{
3     public static void main(String args[]){
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] a = new int[n];
7         for(int i = 0;i<n;i++){
8             a[i]=sc.nextInt();
9         }
10        int min = a[0];
11        for(int i = 1; i<n; i++){
12            if (a[i]<min)
13                min= a[i];
14        }
15        System.out.println(min);
16    }
17 }
```

INPUT

3

10 20 30

OUTPUT

10

29. 2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class secmin{
3     public static void main(String args[]){
4         Scanner sc= new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] a= new int[n];
7         for (int i=0; i<n;i++){
8             a[i]=sc.nextInt();
9         }
10        int small,sec;
11        if (a[0]<a[1]){
12            small=a[0];
13            sec=a[1];
14        }
15        else{
16            small=a[1];
17            sec=a[0];
18        }
19        for (int i=2;i<n;i++){
20            if(a[i]<small){
21                sec=small;
22                small=a[i];
23            }else if (a[i]< sec){
24                sec=a[i];
25            }
26        }
27        System.out.println(sec);
28    }
29 }
```

INPUT

3

10 20 30

OUTPUT

20

30.SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] a = new int[n];
7         for (int i = 0; i < n; i++)
8             a[i] = sc.nextInt();
9         for (int i = 0; i < n - 1; i++)
10             for (int j = 0; j < n - 1 - i; j++)
11                 if (a[j] > a[j + 1]) {
12                     int t = a[j];
13                     a[j] = a[j + 1];
14                     a[j + 1] = t;
15                 }
16         for (int x : a)
17             System.out.print(x + " ");
18     }
19 }
```

INPUT

3

15 13 12

OUTPUT

12 13 15

31. 2ND MIN

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int a[] = new int[n];
8
9         for (int i = 0; i < n; i++)
10             a[i] = sc.nextInt();
11
12         for (int i = 1; i < n; i++) {
13             int key = a[i], j = i - 1;
14             while (j >= 0 && a[j] > key) {
15                 a[j + 1] = a[j];
16                 j--;
17             }
18             a[j + 1] = key;
19         }
20
21         for (int x : a)
22             System.out.print(x + " ");
23     }
24 }
```

INPUT

3

30 10 20

OUTPUT

10 20 30